

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech

Branch : Computer Engineering / Computer Science and Engineering

Semester :V

Subject Code & Name: BTCOC501 Database Systems

Max Marks: 60

Date:01-01-24

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) We develop database applications directly on top of file systems. But some problems arise because of this, Discuss those problems.	CO1/3	6
B) Explain two tier and three tier architectures of DBMS with schematic diagram and mention at least 1 example of each.	CO1/2	6
C) Draw the symbols of ER Diagram. Draw ER Diagram for Library Management System.	CO1/2	6
Q.2 Solve Any Two of the following.		12
A) What is Relational algebra? Explain basic operations along with symbols of Relational Algebra.	CO2/2	6
B) Write Relational Algebra Queries for following. Note: Consider Relational Table - Account(Acno, AcName, Br_name, Amount). a) To display all tuples from Accounts table. b) To display Acno, AcName from accounts. c) To display the tuples for the account whose amount>25000 and Br_name="Dharashiv) d) To display Acno, AcName, Amount From Account where the amount>50000	CO2/3	6
C) Explain Equijoin, left join and right join with suitable example and write proper relational algebraic query for each of them.	CO2/2	6
Q. 3 Solve Any Two of the following.		12
A) Mention different SQL languages. Explain their purpose with suitable SQL commands.	CO3/1,2	6
B) Write the SQL Queries for the following.	CO3/3	6

- a) Creating, Using, showing and deleting databases.
 b) To create BOOK table with Bkid, Title, Author, branch, cost and date of publishing. (Assume proper data type for each column).
 c) Write query to insert 2 valid rows of values into the BOOK table.
 d) Write query to increase the length of Title and Author.
 e) Write query to list books with cost less than 100.
 f) Write query to list all books with title "RDBMS".
- C) Consider the following schema: CO3/3 6
- Suppliers (sid : integer, sname : string, address : string)
 Parts (pid : integer, pname : string, color : string)
 Catalog (sid : integer, pid : integer, cost : real)
- Write query for the following
- a) To create above tables
 b) Find the name of suppliers who supply some red parts
 c) Find the sids of suppliers who supply some red or green parts
 d) Find the sids of suppliers who supply some red part or are at 221 packer Ave
 e) Find the sids of suppliers who supply some red part and some green part
- Q.4 Solve Any Two of the following. 12
- A) What is normalization? Why one need to normalize the database tables? CO4/2 6
 B) Explain 1st, 2nd and 3rd normal form with the help of suitable example. CO4/2 6
 C) Why we need to organize database files? Mention the type of organisations, and explain any one with suitable example. CO3/2 6
- Q. 5 Solve Any Two of the following. 12
- A) Define Transaction and their properties with suitable example. CO5/2 6
 B) Define serial and non-serial schedules. Explain serialisability with suitable example. CO5/2 6
 C) Explain the need of concurrency control. Mention the concurrency control methods. Explain any one with suitable example. CO5/2 6

*** End ***